

Performance Testing

Date	04-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing (streamlit)
Type of Testing	Functional testing performance testing
Target Module / API	Crop prediction module (ML Model)
Test Environment	Local system (windows, Streamlit)
Test Date	02/07/2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Streamlit application	1	10	Application loads successfully without errors
2	Submit valid crop input values	1	20	Correct crop recommendation is displayed
3	Multiple prediction requests	5	30	Application responds correctly without crashing
4	Invalid/empty input validation	1	15	Error message or validation prompt is shown

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Response within target
2	Response Time (Max)	< 5 seconds	2.3 sec	Pass	Stable performance
3	Throughput (Req/ sec)	10 Req/ sec	12 Req/ sec	Pass	Good request handling
4	Error Rate	< 1 %	0 %	Pass	No errors observed
5	CPU Utilization	< 80 %	35 %	Pass	CPU usage normal
6	Memory Utilization	< 80 %	42 %	Pass	Memory usage within limit

Step 4: Observations & Analysis

Key Findings:

> The application responded quickly under normal usage. Crop prediction and recommendation features worked correctly with low response time and stable resource utilization.

Bottlenecks Identified:

> No major bottlenecks were identified. Slight delay may occur during the first model loading.

Optimization Steps Taken:

> Optimized model loading, reduced unnecessary computations, and used efficient data processing with Pandas and NumPy for better performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6



out put):

